

Foundation (Jalapeno)

- Q1.** The graph of the inequality $y > 2x - 1$ has a
 (A) solid boundary line
 (B) dashed boundary line
 (C) no boundary line
 (D) vertical boundary line
 (E) horizontal boundary line
- Q2.** Which inequality is represented by the graph?
 (A) $x + y \leq 4$
 (B) $x + y \geq 4$
 (C) $x - y \leq 4$
 (D) $x - y \geq 4$
 (E) $y \leq -x + 4$
- Q3.** The solution region for the inequality $y < -x + 2$ is
 (A) below the line $y = -x + 2$
 (B) above the line $y = -x + 2$
 (C) on the line $y = -x + 2$
 (D) between the lines $y = -x + 2$ and $y = x - 2$
 (E) outside the lines $y = -x + 2$ and $y = x - 2$
- Q4.** To graph the inequality $y \geq 3x - 2$, first graph the line
 (A) $y = 3x - 2$ as a solid line
 (B) $y = 3x - 2$ as a dashed line
 (C) $y = -3x + 2$ as a solid line
 (D) $y = -3x + 2$ as a dashed line
 (E) $y = 2x - 3$ as a solid line
- Q5.** Which of the following points is a solution to the inequality $x + 2y > 4$?
 (A) (0, 0)
 (B) (0, 2)
 (C) (2, 0)
 (D) (1, 1)
 (E) (3, 1)
- Q6.** The inequality $y \leq -2x + 1$ has a boundary line with a slope of
 (A) -2
 (B) 2
 (C) -1
 (D) 1
 (E) 0
- Q7.** The system of inequalities $y > x - 2$ and $y < -x + 1$ has a solution region that is
 (A) a triangle
 (B) a quadrilateral
 (C) a pentagon
 (D) a hexagon
 (E) unbounded
- Q8.** The inequality $y > x^2 - 4$ is a
 (A) linear inequality
 (B) quadratic inequality
 (C) cubic inequality
 (D) quartic inequality
 (E) rational inequality

Open Ended Questions (FRQ)

- Q9.** Graph the inequality $y \geq 2x - 3$ and shade the solution region. Label the x - and y -intercepts.
- Q10.** Solve the system of inequalities $x + y > 2$ and $x - y < 1$. Graph the solution region and label the corner points.

Chef's Tips

- Tip 1: Emphasize the importance of dashed versus solid boundary lines.
- Tip 2: Have students check their graphs by plugging in test points.
- Tip 3: Use different colors to distinguish between different inequalities.